MINI PROJECT DOCUMENTATION

AI-Based Symptom Analysis and Healthcare Guidance System

A responsible, web-based prototype for symptom awareness and general healthcare guidance

Live Project	organic-smart-health-pulse.base44.app
Platform	Base44 AI application builder
Project Area	Web Development • AI/ML Concepts • Digital Health Awareness

Submitted by	Name: _____
Student Details	Roll Number: _____
Academic Details	B.Tech – 2nd Year, CSE (Artificial Intelligence & Machine Learning)
Institution	College: _____
Project Guide	Name: _____

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Documentation Note

This document describes the live project at the URL supplied by the student. The feature descriptions and technology notes are based on the public pages available during preparation. The application identifies itself as a prototype and states that its current analysis engine uses transparent rule-based logic; it does not claim to use a trained machine-learning model or provide a clinically validated diagnosis.

Project link: https://organic-smart-health-pulse.base44.app

Before final submission, the student should fill in the cover-page details and verify that every feature described here still matches the deployed application.

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1. Abstract

The AI-Based Symptom Analysis and Healthcare Guidance System, presented as HealthPulse AI, is a web-based educational prototype designed to help users organize self-reported symptoms and access general health information. Users can choose an age group, select one or more symptoms, specify duration and severity, add optional context, and answer questions about warning signs. The application is intended to provide general information and guidance about when professional medical attention may be appropriate.

The deployed project was created using Base44, an AI-assisted application-building platform. The project's public About page describes a responsive interface built with React and Tailwind CSS, Base44-supported server logic, and optional consent-based database storage. The current assessment engine is described as a transparent rule-based prototype; no machine-learning model is stated to have been trained. The application emphasizes safety by presenting emergency warnings and making clear that it is not a substitute for a qualified healthcare professional.

Keywords: Healthcare guidance, symptom assessment, Base44, web application, rule-based system, AI/ML project, health awareness, responsible technology.

2. Introduction

Digital tools can make general health information easier to access, but online information can be confusing and symptoms can have many possible causes. A person may not know how to describe symptoms clearly or recognize when urgent medical care is needed. A carefully designed guidance website can help users structure the information they report, read general educational material, and understand that professional evaluation may be necessary.

HealthPulse AI demonstrates how a web application can combine a guided form, rule-based processing, health education content, and safety notices. The project is developed as an academic prototype rather than a clinical decision-support product. It focuses on usability, responsible messaging, and clear limitations.

2.1 Background

The project brings together web application design and introductory AI/ML concepts. A rule-based engine applies defined conditions to the inputs rather than learning patterns from a training dataset. This provides a transparent foundation that can be tested and improved, while making it important not to describe the current implementation as a trained machine-learning model.

2.2 Motivation

The motivation is to improve health awareness, make common symptom information easier to navigate, and encourage timely professional care when warning signs are present. The interface is intended to be understandable to users with different levels of technical knowledge.

3. Problem Statement and Proposed Solution

3.1 Problem Statement

People may struggle to understand common symptoms, locate reliable health information, and judge when to consult a healthcare professional. General search results can be difficult to interpret, and symptom information alone cannot confirm a disease. The project addresses the need for a simple, safety-conscious website that organizes symptom inputs and presents non-diagnostic health information with appropriate caution.

3.2 Proposed Solution

The proposed system offers a guided symptom assessment workflow. Users enter their age group, select symptoms, indicate how long they have had them and how severe they are, optionally add information, and answer questions about potential warning signs. The system processes the submitted answers using its configured rule-based logic and presents general health information and recommended next steps. Emergency guidance is displayed for potentially serious warning signs.

3.3 Expected Value

- Provides a structured way to record the symptoms a user reports.
- Makes general health education easier to browse.
- Highlights warning signs and encourages appropriate medical attention.
- Demonstrates the use of an AI application builder to develop a working web prototype.
- Promotes transparency by explaining that outputs are informational rather than confirmed diagnoses.

4. Objectives and Scope

4.1 Objectives

- Develop a clear, responsive, and accessible healthcare guidance website.
- Collect symptom selections, duration, severity, and relevant warning-sign responses.
- Process user inputs with transparent application logic.
- Present general health information and suitable next-step guidance.
- Give priority to urgent safety messages when red-flag symptoms are reported.
- Explain privacy, limitations, and the need for professional medical care.

4.2 Scope

The scope includes a home page, symptom assessment form, general health information library, project information, help/settings, privacy/disclaimer information, and a dashboard entry point. The assessment is educational and non-diagnostic. The project does not claim to perform laboratory analysis, examine a patient, confirm a disease, prescribe medicine, or replace emergency services.

4.3 Intended Users

The intended audience is people seeking general health information and students demonstrating responsible digital-health application design. The tool should not be relied on for emergency decisions or used as a substitute for advice from a qualified clinician.

5. System Requirements

5.1 Functional Requirements

ID	Requirement	Expected behaviour
FR-01	Home page	Explains purpose, provides navigation, and displays a safety disclaimer.
FR-02	Symptom input	Allows an age group, multiple symptoms, duration, severity, and optional details.
FR-03	Warning-sign questions	Collects yes/no answers for listed warning signs.
FR-04	Assessment processing	Uses submitted input to generate general, non-diagnostic guidance.
FR-05	Health library	Provides searchable educational content and warning-sign information.
FR-06	Project information	Explains objectives, technologies, analysis approach, and limitations.
FR-07	Privacy/disclaimer	Explains intended use, limits, and data-handling principles.
FR-08	Dashboard/history	Provides a dashboard entry point; saved records should only be used with consent and access controls.

5.2 Non-Functional Requirements

- **Usability:** Forms and results should use clear labels and plain language.
- **Responsiveness:** Pages should adapt to desktop, tablet, and mobile screens.
- **Reliability:** Invalid or incomplete input should be handled without crashing.
- **Privacy:** Collect only necessary information; save assessment history only with consent.
- **Safety:** Emergency warnings must be prominent and must not be delayed by optional analysis.
- **Maintainability:** Symptom mappings and educational content should be easy to review and update.

5.3 Hardware and Software Requirements

Category	Minimum / Suggested requirement
Client device	Laptop, desktop, tablet, or smartphone with a modern web browser.
Internet	Internet connection to access the deployed Base44 application.
Browser	A current version of Chrome, Edge, Firefox, or Safari.
Development platform	Base44 AI application builder.
Frontend technologies	React and Tailwind CSS, as described on the project's About page.
Backend/data	Base44-supported server logic and database features, where enabled.

6. System Architecture

The application follows a web-based flow. A user interacts with the frontend, submits structured responses, and receives results from the application's processing logic. The public About page describes the interface as React and Tailwind CSS, backend processing as Base44-supported server logic, and data storage as consent-based.

USER
Select symptoms • duration • severity • warning-sign answers
↓
FRONTEND WEB INTERFACE
React + Tailwind CSS (Base44-generated UI)
↓ **submit form**
BASE44-SUPPORTED APPLICATION LOGIC
Validate input and apply transparent rule-based conditions
↓
GUIDANCE AND SAFETY OUTPUT
General health information • next steps • warning messages
↕ **optional consent-based storage**
DATABASE / ASSESSMENT HISTORY (IF ENABLED)

6.1 Data Flow

- The user opens the website and reads the purpose and disclaimer.
- The user selects an age group, one or more symptoms, duration, and severity.
- The user answers the listed warning-sign questions and submits the form.
- The application processes the inputs through the configured rule-based assessment logic.
- The results interface presents general information and safety-oriented next steps.
- If history storage is available and the user consents, the application may save a record subject to privacy controls.

6.2 Analysis Approach

The current public project description states that the assessment engine uses transparent rule-based logic derived from public health guidance. A rule-based system applies conditions written by the developer; it is not the same as a trained ML model. An AI integration or validated model could be considered in future work, but it must be evaluated, constrained, and prevented from overriding emergency safety rules.

7. Methodology and Working Process

7.1 Development Method

The project follows an iterative prototype workflow: define the problem, generate the application using Base44, review the generated interface, connect and test the assessment logic, check safety messages, and refine the pages. Each iteration should be tested with sample inputs before further features are added.

7.2 User Workflow

Step	User/System action	Outcome
1	Open the home page.	User sees purpose and disclaimer.
2	Choose age group and symptoms.	Symptoms are captured in structured form.
3	Select duration and severity.	Additional context is included.
4	Answer warning-sign questions.	Potential urgent indicators are captured.
5	Submit the assessment.	Application validates and processes the data.
6	Read the results.	User receives general information and next-step guidance.

7.3 Example Demonstration

Illustrative input: symptom = headache; duration = 1–2 days; severity = mild; no warning signs reported. The application should summarize these answers and display general educational information, appropriate self-care information where supported by its content, and signs that should prompt medical attention. This example is not a diagnosis and does not establish that the user is medically safe.

8. Modules and Features

Module	Description
Home	Introduces HealthPulse AI, explains the purpose, and provides navigation to assessment and information pages.
Symptom Assessment	Offers age-group selection, searchable symptom choices, duration and severity options, optional notes, and warning-sign questions.
Analysis and Guidance	Processes entered information and provides general non-diagnostic guidance. The current project description identifies a rule-based prototype.
Health Information Library	Contains educational information on common symptoms, general wellness, emergency warning signs, and preparing for a doctor appointment.
Dashboard	A navigation destination for dashboard/history features. Confirm actual behaviour in the deployed app before demonstrating saved records.
About the Project	Explains the problem statement, objectives, role of AI/ML, technology choices, and limitations.
Help & Settings	Provides help and settings navigation; availability of individual controls should be checked in the deployed version.
Privacy & Disclaimer	Explains informational use, limitations, privacy considerations, and emergency guidance.

9. Technology Stack

Technology / component	Role in the project
Base44	AI-assisted application-building platform used to create and deploy the web application.
React	Frontend component framework identified on the project's About page.
Tailwind CSS	Utility-first styling approach identified on the project's About page.
Base44 server logic	Processes submitted symptom information and supports application behaviour.
Base44 database	Described as available for assessment records when a user consents to saving them.
Rule-based logic	Current described assessment approach; applies predefined conditions to user responses.
Web browser	Allows users to access the deployed application without installing a separate desktop program.

9.1 Why Base44?

Base44 reduces the amount of manual setup required to create a web application by helping generate pages, UI components, application logic, and deployment. This makes it useful for a student prototype that needs a working interface quickly. The student should still review generated behaviour, validate data handling, test navigation, and document which features are actually implemented.

9.2 Role of AI/ML

The use of an AI-assisted development platform does not automatically mean the finished product contains a trained AI/ML model. According to the deployed project's public About page, this version uses a rule-based assessment engine and has no trained machine-learning model. This is an important distinction for an academic presentation. The current project demonstrates AI-enabled development and rule-based health guidance; model training and validation are possible future extensions.

10. Testing and Validation

Testing should confirm that the user interface, form validation, assessment flow, safety messages, and information pages behave as expected. The following test cases are a recommended checklist for the student to execute on the live site. They are not a claim that every test has already passed.

Test ID	Test case	Expected result
TC-01	Open the home page.	Page loads and displays project purpose and disclaimer.
TC-02	Open assessment page.	Symptom form and input categories are visible.
TC-03	Select multiple symptoms.	Multiple selected symptoms are retained in the form.
TC-04	Submit with required fields missing.	The form requests missing information or prevents invalid submission.
TC-05	Submit valid sample inputs.	Results reflect the information actually submitted.

Test ID	Test case	Expected result
TC-06	Answer Yes to a listed emergency warning sign.	Prominent emergency guidance appears; routine guidance does not delay it.
TC-07	Open Health Information Library.	Educational content and warning-sign guidance are readable.
TC-08	Open About and Disclaimer pages.	Project purpose, limitations, and non-diagnostic status are explained.
TC-09	View on a mobile-sized screen.	Content remains readable and controls remain usable.
TC-10	Check assessment history/storage, if enabled.	Records are private, consent-based, and deletable as stated.

Testing note: Record the actual outcome of each test during your demonstration. Do not mark a test as passed unless you have run it on the deployed application.

11. Privacy, Safety and Ethical Considerations

11.1 Medical Safety

- The application provides general information and must not claim a confirmed diagnosis.
- It must not prescribe medication or personalized dosages.
- Emergency messages should be visible and immediate when a listed warning sign is selected.
- The site tells users in India to call 112, or their local emergency number, for severe emergency warning signs.
- Users should seek qualified medical care for urgent, worsening, or concerning symptoms.

11.2 Privacy and Data Protection

- Collect only the information needed for the stated purpose.
- Clearly explain whether symptom information is stored and why.
- Save assessment history only when consent is obtained, as described by the project.
- Restrict saved records so users cannot view another user's private information.
- Provide deletion controls where stored records are supported.
- Avoid including sensitive symptom details in public URLs, logs, or public-facing pages.

11.3 Responsible Use

Health information is sensitive and can affect important decisions. The interface should communicate uncertainty, use trustworthy references, avoid alarming or falsely reassuring language, and be reviewed regularly. The project is an academic prototype and has not been clinically validated. It should not be presented as a medical device or a substitute for a clinician.

12. Limitations and Future Enhancements

12.1 Current Limitations

- A symptom form cannot capture the full context of a medical examination.
- User inputs may be incomplete, mistaken, or ambiguous.
- Different conditions can cause similar symptoms.
- Rule-based mappings cover only the cases explicitly described in the logic.
- The deployed project states that it is not clinically validated and does not use a trained ML model.
- Language support, saved history, and dashboard behaviour should be described only to the extent they are fully implemented and tested.

12.2 Future Enhancements

- Add professionally reviewed and versioned health information from authoritative sources.
- Evaluate any future ML model on suitable, ethically sourced data with expert review and documented performance.
- Add Telugu, Hindi, Kannada, and Tamil translations with medical-language review.
- Add text-to-speech or audio guidance for accessibility.
- Improve consent, account security, and user-controlled deletion for saved assessment history.
- Create a test dashboard to record functional test results and known limitations.
- Add feedback collection that avoids requesting unnecessary sensitive medical details.

13. Conclusion

The AI-Based Symptom Analysis and Healthcare Guidance System, branded as HealthPulse AI, is a web-based mini project that demonstrates how a guided symptom form, rule-based processing, health education, and safety messaging can be combined into a digital health-awareness tool. Base44 was used as the AI-assisted application-building platform, and the live project describes a React and Tailwind CSS interface with Base44-supported backend logic.

The project's main contribution is a structured and accessible way to present general information and encourage appropriate healthcare-seeking behaviour. Its limitations must remain clear: symptom selections do not confirm a disease, the prototype is not clinically validated, and the current description does not claim a trained machine-learning model. Further development should prioritize medically reviewed content, rigorous testing, privacy, accessibility, and safety.

13.1 Project URL

https://organic-smart-health-pulse.base44.app

14. References

No.	Reference
1	HealthPulse AI – live project website. https://organic-smart-health-pulse.base44.app
2	HealthPulse AI – About the Project page. https://organic-smart-health-pulse.base44.app/about
3	HealthPulse AI – Symptom Assessment page. https://organic-smart-health-pulse.base44.app/assessment
4	HealthPulse AI – Health Information Library. https://organic-smart-health-pulse.base44.app/library
5	World Health Organization (WHO). https://www.who.int/
6	MedlinePlus, U.S. National Library of Medicine. https://medlineplus.gov/
7	NHS Health A to Z. https://www.nhs.uk/conditions/
8	Centers for Disease Control and Prevention (CDC). https://www.cdc.gov/

Appendix A – Suggested Live Demonstration

- Open the home page and explain the project's purpose and disclaimer.
- Open Symptom Assessment and select a sample symptom, duration, and severity.
- Show the warning-sign questions and explain why emergency safety takes priority.
- Submit a non-emergency illustrative example and review the generated guidance.
- Open the Health Information Library and show its general education content.
- Open About the Project and explain that the current engine is rule-based, not a trained ML model.
- Conclude by discussing limitations, privacy, and possible future enhancements.

IMPORTANT: This project is an educational prototype, not a clinically validated diagnostic system. In an emergency, call 112 in India or your local emergency number.